2.2.2 Length of the arc and Area of the sector

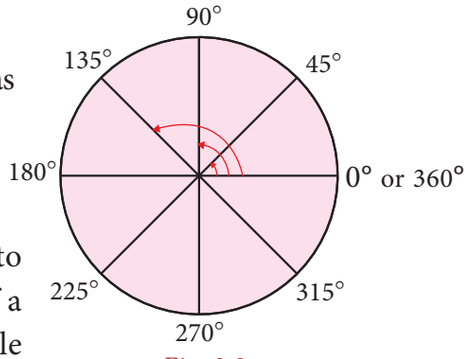


Fig. 2.8

We have already learnt that a circle of radius '*r*' *units* has

Central angle = 360°

Circumference of the circle = $2\pi r$ *units*

Area of the circle = πr^2 *Sq.units.*

First, let us take a circle . If a circle is divided into 2 equal sectors we will get 2 semi-circles. The length of a semi-circular arc is half of the circumference of the circle and the area of the semi-circle is half of the area of the circle.

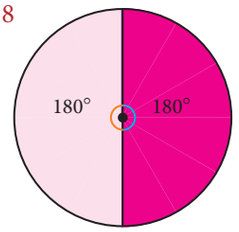


Fig. 2.9

Length of the semi-circular arc $= \frac{1}{2} \times 2\pi r = \frac{180^\circ}{360^\circ} \times 2\pi r$ *units.*

Area of the semi-circle $= \frac{1}{2} \times \pi r^2 = \frac{180^\circ}{360^\circ} \times \pi r^2$ *sq.units.*

Smilarly, if a circle is divided into 3 equal sectors,

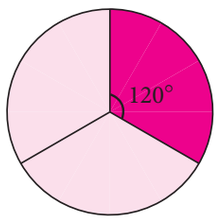


Fig. 2.10

Length of the arc of this sector $= \frac{1}{3} \times 2\pi r = \frac{120^\circ}{360^\circ} \times 2\pi r$ *units*

Area of the sector $= \frac{1}{3} \times \pi r^2 = \frac{120^\circ}{360^\circ} \times \pi r^2$ *sq. units.*

In the same way, if a circle is divided into 4 equal sectors,

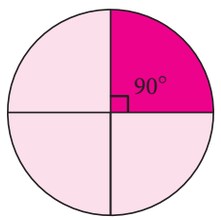


Fig. 2.11

Length of the arc of the circular quadrant $= \frac{1}{4} \times 2\pi r = \frac{90^\circ}{360^\circ} \times 2\pi r$ *units.*

Area of the quadrant $= \frac{1}{4} \times \pi r^2 = \frac{90^\circ}{360^\circ} \times \pi r^2$ *sq.units*

What do we know from this?

If the ratio of the central angle of a sector to the central angle of a circle is multiplied with the circumference and the area of the circle, we can find the length of the arc of that sector and its area respectively.

That is, if we assume that the central angle of a sector of radius '*r*' *units* as θ°, then, the ratio of the central angle θ° to 360° is $\frac{\theta^\circ}{360^\circ}$.

Length of the arc, $l = \frac{\theta^\circ}{360^\circ} \times 2\pi r$ *units*

Area of the sector, $A = \frac{\theta^\circ}{360^\circ} \times \pi r^2$ *sq.units.*

Think



Instead of multiplying by $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$, we shall multiply by $\frac{180^\circ}{360^\circ}$, $\frac{120^\circ}{360^\circ}$ and $\frac{90^\circ}{360^\circ}$ respectively. Why?

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